Biomolecular Sciences

Vrije Universiteit Amsterdam - Faculty of Science - Biomolecular Sciences - 2025-2026

Modern society faces a number of demanding challenges going forward. Safe, stable food & crop production and developing novel treatments for diseases in aging or infection are priorities. Next to this, we also need to find novel solutions for emerging problems, such as broad antibiotic resistance or nitrogen fixation in our soils.

With a thorough understanding of molecular and cellular processes and a focus on fundamentals, we will be better positioned to innovate and develop biotechnological solutions or treatments to these issues. Moreover, today's biomolecular scientists are increasingly exposed to huge volumes of data from high-throughput methods which require a critical scientific approach. To this end, it is vital that molecular and cell biology graduates are well-prepared to enter the lab within an interdisciplinary Life Sciences research community.

Teaching in the program takes the form of lectures, research tutorials, work groups, demonstrations, lab visits, lab and computer practicals, as well as onsite training and mentoring.

In the first year you will follow the obligatory courses Protein Science, Fundamentals of Bioinformatics, From Genome to Function and Cell Structures and Function in the first two periods. In period 3 you will start to tailor your program to you yourself by choosing an optional course. An internship (see below) of five months will complete your first year.

In the second year you will learn about the Lifecycle of Research Projects and do more courses of your own choice. We offer sets of courses along a specific theme, but here you will also have the opportunity to combine courses from different themes. You complete your program with a second internship or bioinformatics project.

In the program you will do two large research projects (internships). One is always an experimental (wetlab) internship and the second project can be another experimental (wetlab) internship or a Bioinformatics/Data analysis project. These projects can be performed under the mentorship and guidance of a researcher of your choice. We also encourage our students to consider going abroad or in industry for their internship. Our lecturers can help you with finding a suitable lab through their own extensive research networks.

In addition, over the two years, you will follow a series of workshops on strengthening your academic skills and preparing yourself for the job market.

Some examples of current research topics in the biomolecular field

· A new method to fight tuberculosis

After a period of relative calmness, tuberculosis seems to be on the advance again, causing currently some 2 million casualties per year. *Mycobacterium tuberculosis* excretes different proteins all playing a role in disease development. Knowledge about the underlying mechanisms of excretion can be applied in the development of new drugs to fight the bacteria.

· Regulation of gene expression

It is a miracle how a single fertilized egg cell develops into a complex organism and how undifferentiated stem cells can be assigned to take their role in this process. Expression and regulation of genes in eukaryotes is controlled by transcription factors, but in addition there is also post-transcriptional control. Unravelling these regulatory networks will open up huge possibilities in biomedical and biotechnological research.

· Virus-induced cancer

Do viruses play a role in the development of cancer? Research from the VU University in Amsterdam suggest that they indeed do. These findings are important for the development of new cancer therapies.

· Busy traffic within cells

Within our cells, there is ongoing and busy traffic of nutrients, organelles and signalling compounds, driven by motor proteins. In order to visualize, count and track these motor proteins and figure out how they cooperate in their job advanced new microscopy methods have been and still need to be developed.

Info

Level	Master
Language	English
Duration	2 years
Type	Full Time

Credits	120 EC
Faculty	Faculty of Science

Teaching and examination regulation 2025-2026
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Biomolecular Sciences year 1

Description

In the first year you will follow the obligatory courses Protein Science, Fundamentals of Bioinformatics, From Genome to Function and Cell Structures and Function in the first two periods. In period 3 you will start to tailor your program to your yourself by choosing a course or do a literature review in Topics in Biomolecular Sciences. You will complete year 1 with one of the two end projects, which is the first internship.

In this year you will also do the first part of Professionalism in Biomolecular Sciences.

Programme Sections

- Biomolecular Sciences year 1 compulsory courses

Biomolecular Sciences year 1 compulsory courses

Courses

Course Name	Period	Credits	Code
Internship M BMOL I	Ac. Year (September)	30EC	AM_1294
Professionalism in Biomolecular Science year 1	Ac. Year (September)	0EC	AM_1273
Fundamentals of Bioinformatics	P1	6EC	X_405052
Protein Science	P1	6EC	AM_470145
Cell Structures and Functions	P2	6EC	AM_470615
From genome to function	P2	6EC	AM_1290

Biomolecular Sciences year 2

Description

In the second year you will learn about the Lifecycle of Research Projects and do courses of your own choice. We offer sets of courses along a specific theme, but here you also have the opportunity to combine courses from different themes. You complete your program with a second internship, that can also be done on a bioinformatics topic.

Programme Sections

- Biomolecular Sciences year 2 compulsory courses

Biomolecular Sciences year 2 compulsory courses

Courses

Course Name	Period	Credits	Code
Internship M BMOL II	Ac. Year (September)	33EC	AM_1306
Professionalism in Biomolecular Science year 2	Ac. Year (September)	3EC	AM_1277
Lifecycle of Research Projects	P1, P1	6EC	AM_1293

Biomolecular Sciences year 1 or 2 compulsory courses

Courses

Course Name	Period	Credits	Code
Topics in Biomolecular Sciences	Ac. Year (September)	6EC	AM_1311

Biomolecular Sciences year 1 and 2 electives

Description

To complete your 120ECs, you can choose three courses from this section. Some courses are clustered into themes, but you are allowed to mix and match between themes. You have to check yourself whether the schedules do not collide.

Themes:

- **(Systems) Biology:** Introduction to Systems Biology, Molecular Infection Biology, The Human Microbiome in Health and Disease, Basic Models of Biological Networks
- **(Chemical) Molecules and Toxicology:** Principles of Drug Targets, Molecular and Cellular Toxicology, Drug Target Biochemistry and Signaling
- **Bioinformatics:** Algorithms in Sequence Analysis, Biosystems Data Analysis, Structural Bioinformatics, Bioinformatics for Translational Medicine
- **Tools & Analysis:** Biophotonics, Statistics with R, 3Rs of animal experiments applying human cells in culture, Quantitative Single-cell Biology, Advanced Biophysics

Courses

Course Name	Period	Credits	Code
Introduction to Systems Biology	P1	6EC	X_428565
Oncogenesis	P1	6EC	M_OONC03
Principles of Drug Targets	P1	6EC	XM_0105
Algorithms in Sequence Analysis	P2	6EC	X_405050
Drug Target Biochemistry and Signaling	P2	6EC	XM_0111
Molecular and Cellular Toxicology	P2	6EC	XM_0113
Molecular Infection Biology	P2	6EC	AM_470657
Statistics with R	P2	6EC	X_418156
Biophotonics	P3	6EC	AM_1213
Biosystems Data Analysis	P3	6EC	XM_0078
The Human Microbiome in Health and Disease	P3	6EC	AM_1310
Basic Models of Biological Networks	P4	6EC	X_418154
Structural Bioinformatics	P4	6EC	X_405019
Advanced Biophysics	P5	6EC	X_422583
Bioinformatics for Translational Medicine	P5	6EC	X_405092
Replacement, Refinement and Reduction of animal experiments applying human cells in culture	P5	6EC	XM_0148
Quantitative Single-cell Biology	P6	6EC	XM_0084

Courses

Advanced Biophysics

Course Code	X_422583
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Dulin
Examiner	dr. D. Dulin
Teaching Staff	dr. D. Dulin, dr. ir. I. Heller
Teaching method(s)	Lecture

Course Objective

Learning objectives

- Understand the physics underlying the biological problems and biophysical research methods featured in this course
- Be able to explain how these methods can be exploited to address specific biological problems and identify strengths and weaknesses of methods in biological context
- Critically reflect on your understanding of methods application in assigned literature and formulate relevant questions
- Analyze scientific literature that uses the discussed methodologies to addresses biological problems and reflect on suitability of used methods
- Communicate and present your literature analysis and critically reflect on analyses by other students

Course Content

Life is, by its very definition, a dynamic quantity. In this course an overview is given of current biophysics, biophotonics, and biomedical research fields that investigate the dynamic processes that take place in biomolecules, membranes, cells and tissues and are indispensable for biological functionality. This course aims to highlight the 'what', 'why' and 'how' of these research fields: what are the established and new biophysical insights, why are these insights important and interesting, and how are these results being obtained.

The following biophysical/biomedical topics and techniques are featured in this course:

- DNA processing and dynamics
  o Topics: DNA structure and stability, DNA/RNA polymerase, DNA architectural proteins, DNA repair.
  o Techniques: optical tweezers, AFM, tethered particle motion, magnetic tweezers, superresolution imaging.
- Protein dynamics
  o Topics: Photoactive proteins, light-driven enzymes, Motor proteins, optogenetics.
  o Techniques: infrared spectroscopy, time-resolved spectroscopy, laser techniques
- Membrane dynamics
  o Topics: Photosynthesis, membrane protein diffusion, Neuroimaging.
  o Techniques: AFM, single molecule, nonlinear imaging, polarized spectroscopy.
- Cell/tissue dynamics
  o Topics: Optical and/or mechanical characterization of tissue or cells.
  o Techniques: (Fluorescence) microscopy, optical coherence microscopy, mechanical analysis.

Additional Information Teaching Methods

Lectures, guest lectures, literature essay and assignments, oral literature presentation

Method of Assessment

- Literature assignments
- Essay (literature or research proposal) (70% final grade)
- Essays from two other students peer review (10% final grade)
- Oral literature presentation (20% final grade)

Entry Requirements

BSc. Physics, BSc. Medical Natural Sciences, BSc Chemistry or comparable.

For MNS students to participate in this course, it is advised to have successfully completed the course Microscopie en spectroscopie voor MNW (X_420586) and Mechanics and Thermodynamics in the Cell (X_422589).

Literature

Notes, handouts and papers.

Additional Information Target Audience

mMNS-PoL, mPhys-LSBP, mPhys-PLH

Algorithms in Sequence Analysis

Course Code	X_405050
Credits	6
Period	P2
Course Level	600
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	R. Kumar
Examiner	dr. S. Pissis
Teaching Staff	dr. J.M. Urtasun Elizari, dr. ir. K.A. Feenstra, dr. S. Pissis, prof. dr. J. Heringa, dr. Z. Işik MSc, H. Verbeek MSc
Teaching method(s)	Computer lab, Lecture

Course Objective

Have you ever wondered how we can track a gene across 3 billion years of evolution? Or how you can use the genome information of a given cancer patient to find out what may be wrong? Sequence alignment can be used to compare genomes, genes or proteins from bacteria all the way to humans, while further derived algorithms may be employed to make a phylogeny (to find out about evolutionary relationships), find a functional motif in a protein sequence, or a viral sequence in a genome. In this course we focus on the most important algorithms for biological sequence analysis that can be applied to real scientific problems in biology.

After completing the course successfully,

- You can demonstrate in-depth knowledge about the theory of sequence analysis methods.
- You know how to implement several of the most important sequence alignment and analysis algorithms.
- You understand the mathematical formulation behind sequence analysis algorithms.
- You will have understanding and skills to apply sequence analysis algorithms to protein and DNA sequences.
- You will have hands-on experience in tackling biological problems using sequence analysis algorithms, including the statistical framework of HMMs and algorithms used in genome sequencing and analysis.
- You can decide which algorithm is best suited for a particular biological sequence analysis problem.
- You can read through scientific and technical literature and learn to translate algorithms described in text and formal mathematical notation into computer code.

Course Content

Theory:

- Dynamic programming
- database searching
- pairwise and multiple alignment
- probabilistic methods including HMMs
- pattern matching
- entropy measures
- evolutionary models
- and phylogeny

Practical:

- Programming (in Python) an alignment algorithm based on dynamic programming;
- Aligning sequencing data from tumors to the human genome and analysing structural variants;
- Programming (in Python) an implementation of HMMs and using it to predict protein domain structure.

Additional Information Teaching Methods

13 lectures: 2 two-hour lectures per week.
13 computer practicals and associated assignments: 2 two-hour hands-on sessions per week.

Method of Assessment

The final grade for this course will consist of 50% practical work (see above) and 50% theoretical assessment. (assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

The theoretical assessment will be an oral and/or written exam (depending on number of students).

Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Entry Requirements

Bachelor in any science discipline (including medicine).
Basic programming skills (Python) and an interest in biological problems.

Literature

Course material on Canvas.

Books: Durbin, R., Eddy, S.R., Krogh, A., Mitchison, G.. Biological Sequence Analysis. Cambridge University Press, 1998, 350 pp., ISBN 0521629713.

Recommended reading: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics. Garland Science 2008. ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Artificial Intelligence
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)

We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. Ideally, this laptop supports a command line shell, Python 3 and a text editor with syntax highlighting -- either standalone (e.g. Atom or Sublime Text) or as part of a simple IDE (e.g. Spyder). As such, we recommend the Anaconda python distribution regardless of operating system, along with PuTTy or PowerShell for Windows users specifically.

The course is taught in English.

Basic Models of Biological Networks

Course Code	X_418154
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. F.J. Bruggeman
Examiner	prof. dr. F.J. Bruggeman
Teaching Staff	prof. dr. F.J. Bruggeman, dr. G. Stolf Jeuken
Teaching method(s)	Seminar, Computer lab, Lecture

Course Objective

This course focuses on teaching the creation and analysis of mathematical models for biological systems across various levels of organization, from molecular circuits to cellular systems and potentially ecosystems. The primary modelling formalism explored is ordinary differential equations (ODEs), providing a foundation for understanding biological dynamics. The course aims to equip students with the ability to reproduce models from scientific literature and apply them to solve practical problems. It combines lectures with workgroups involving problem-solving using both computational tools (Python) and pen and paper.

Learning Goals:

- Learn to develop and analyse mathematical models (primarily ODE-based) for diverse biological networks (molecular, cellular, etc.).
- Understand how to represent biological processes, particularly molecular mechanisms like enzyme kinetics and protein regulation, using mathematical formalisms like state-transition diagrams and mass-action kinetics.
- Gain proficiency in simplifying complex mechanisms into net process models suitable for integration into larger system models.
- Develop the skills to reproduce existing models from scientific papers and apply modelling techniques to solve real problems.
- Acquire the intuition necessary to construct useful and informative models by analysing various examples.
- Understand the difference between mechanistic and phenomenological/empirical models and their applications.

Course Content

The course slowly builds on the principles required to make useful models of biological dynamics, starting from first principles:

(1) Fundamentals of Dynamical Systems in Biology:

- Introduction to change, dynamics, systems organization, and the role of energy in biological systems.

(2) Modelling Molecular Circuits:

- Using state-transition diagrams to visualize molecular states and transitions.
- Applying mass-action kinetics to describe the rates of spontaneous molecular reactions and processes within state-transition diagrams.
- Developing mathematical models based on state-transition diagrams.

(3) Net Models and Simplification:

- Modelling enzyme kinetics, including catalytic cycles, rapid-equilibrium approximations, steady-state methods, and cooperative effects.
- Developing equilibrium-binding equations for regulatory proteins (e.g., transcription factors, receptors).
- Reducing complex state-transition diagrams to simpler net process models for use in larger systems.

(4) Models of Larger Systems:

- Applying simplified models to study metabolism, signalling, and signal transduction.
- Introduction to phenomenological and empirical models for systems where detailed mechanisms are unknown or too complex (e.g., cell growth, ecosystems).

(5) Practical Application:

- Solving exercises using Python and pen and paper to reinforce concepts and build modelling skills.

Additional Information Teaching Methods

- Lectures
- Workgroups
- Self-study
- Computer practicals
- Computer modelling tutorials

Method of Assessment

- 80% Written exam
- 20% Computer exam (Python)

You must have a grade higher than 5.5/10 on both exams to pass.

Literature

A course syllabus written by the lecturer

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Mathematics

Bioinformatics for Translational Medicine

Course Code	X_405092
Credits	6
Period	P5
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. E. Bosdriesz
Teaching Staff	dr. E. Bosdriesz, dr. J.M. Urtasun Elizari
Teaching method(s)	Practical, Lecture

Course Objective

Observations from biological high-throughput experiments allow us to improve diagnosis and give a personalized treatment plan for patients. However, integrating data from several sources and using this data for predictions is non-trivial.

This is a theoretical and practical Bioinformatics course on computational methods for Translational Medicine; we will focus on Bioinformatics methods that are used to predict the clinical outcome for patients and analysis methods to obtain deeper understanding of complex diseases, by combining data from various high-throughput experiments such as proteomics, microarrays and next-generation sequencing as well as existing biological databases.

After completing the course successfully,

- You can interpret research results in the context of biology
- You will be aware of Bioinformatics methods that are applicable to the area of Translational Medicine
- You can combine these methods into a creative solution using large-scale biological experiments
- You will have hands-on experience in handling large biological datasets, and will understand the complexity of the biological data both from high-throughput experiments and existing biological databases
- You will be able to work as a team to execute a large, complex biomedical bioinformatics project, and report on the results
- You will be familiar with a few in-depth research topics that lie within the expertise area of several (Bioinformatics) researchers, among others from the VU, UvA, AUMC and NKI

Course Content

Theory

- Computational analysis of molecular profiling techniques, such as WGS, WES, proteomics, RNA sequencing, and arrayCGH.
- Computational methods applied to these data types, such as: machine learning, normalisation, regularization, feature selection, classification, read mapping, clustering.
- All data analysis is relevant in a clinical setting, for diagnosis, treatment decisions or biomarker discovery.

Practical

- A large assignment for which you have to build a classifier based on molecular profiling data (the precise data-set is to be determined). You need to hand in predictions, write a paper and give a presentation. Note that this is a group project, and to participate in this practical you need to have obtained at least 12 ECT from Master-level courses.
- A smaller assignment that involves analysing and understanding molecular profiling data of another modality than the large classifier assignment.

Additional Information Teaching Methods

- 13 lectures (2 two-hour lectures per week)
- 12 computer practicals (2 two-hour sessions per week)

Method of Assessment

The final grade for this course will consist of 50% practical work (see above) and 50% theoretical assessment.

Practical assessment (50%):

- Classifier assignment (45%)
- Other data modality assignment (5%)

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Theoretical assessment (50%):

- Oral or written exam (depending on number of course students)
- The exam is based on a selection of 8-10 scientific papers in the field of Bioinformatics & Translational Medicine.

Entry Requirements

Importantly: to participate in the group assignment, you need to have obtained a minimum of 12 ECT of Master level courses. In addition, some basic programming skills (either in R or Python) are required, as well as some basic knowledge on molecular biology. If you are not following the Master Bioinformatics and Systems Biology, it is

advisable to first follow the Master course "Fundamentals of Bioinformatics".

Literature

- Course material on Canvas, including slides and recordings of lectures.
- 8-10 scientific papers are provided, and make up the course syllabus.

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

The course is taught in English.

- Compulsory course for students in Master Bioinformatics and Systems Biology
- Optional course for students with a Bachelor in Physics, Chemistry, Mathematics, Computer Science, Biology, or Biomedical Sciences (see requirements below).

Recommended background knowledge

If you are not following the Master Bioinformatics and Systems Biology, it is advisable to first follow the Master course "Fundamentals of Bioinformatics".

Biophotonics

Course Code	AM_1213
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. Y.J.M. Bollen
Examiner	dr. ir. Y.J.M. Bollen
Teaching Staff	dr. ir. Y.J.M. Bollen
Teaching method(s)	Practical, Lecture

Course Objective

To introduce students into various spectroscopic and microscopic techniques.
Students should know the theoretical principles and the applicability in life sciences of:

- absorption spectroscopy
- fluorescence spectroscopy
- light microscopy
- fluorescence microscopy

Students should be able to:

- plan and conduct experiments using optical techniques
- evaluate results on the basis of theoretical knowledge and recent literature
- present their results in short reports and one or two journal-style papers

Course Content

Optical spectroscopy and microscopy are widely used in cell biology and biophysics. In this course the principles of many of these techniques, including absorption spectroscopy, various types of fluorescence spectroscopy (e. g. polarization, FRET) and fluorescence microscopy (e. g. confocal, TIRF, lifetime imaging) are explained. Their application in modern biophysics and cell biology research is illustrated by a number of (guest) lecturers.
In addition, students will obtain hands- on experience with absorption spectroscopy, fluorescence spectroscopy (e. g. FRET and anisotropy) and fluorescence microscopy. Small groups of students will prepare the experiments,

discuss them with the lecturer and carry them out. The group will write a short report on each experiment and one or two journal-style papers.

Additional Information Teaching Methods

Lectures (28 hours), group assignment (8 hours), self-study, execution of experiments and data analysis (±24 hours) in small groups.

Method of Assessment

Written exam (individual, 50%), oral presentation by group (10 %),
written report (per group, 40%). All parts need to be graded 5.5 or higher in order to pass the course.

This course follows the VU guidelines on the use of generative AI in education.

Literature

Papers, hand-outs, reader and protocols will be made available online.

Additional Information Target Audience

MSc students Biology, Biomolecular Sciences, Biomedical Sciences, Medical Natural Sciences, Physical Sciences, Chemistry, or related.

Additional Information

Due to largely overlapping contents this course is NOT intended for students who have followed the Minor “Biomedische beeldvorming”.

Biosystems Data Analysis

Course Code	XM_0078
Credits	6
Period	P3
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.A. Westerhuis
Examiner	dr. J.A. Westerhuis
Teaching Staff	dr. D. Molenaar, dr. J.A. Westerhuis, dr. G. Stolf Jeuken
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

The amount of omics data being gathered, analysed, integrated and interpreted increases every year. The number of algorithms, packages and webservices also increases very fast. With the high number of omics features e.g. genes, proteins or metabolites measured, the chances of drawing incorrect conclusions are rather high.

In this course we will discuss properties of the measured data, assumptions of the data analysis methods being used, data preprocessing and pretreatment and validation of the conclusions and interpretations.

Upon completion of this course, students will have reached the following goals

1. Students are able to use R for multivariate analysis of omics data. [KBB 4]
2. Students understand linear algebra concepts that are used in data analysis methods. [KBB 5]
3. Students are able to apply data preprocessing methods (variance stabilization, scaling, transformation), explorative data analysis methods (PCA, Clustering) and supervised data analysis (PCDA, ASCA, Random Forest). [KAR 1]
4. Students are able to evaluate which preprocessing methods, and which data analysis method to use for a given problem. [KAR 5]
5. Students are able to analyse omics data and interpret the results. [KAR 1]
6. Students are able to evaluate the quality of the data analysis models using cross validation and permutation methods. [KAR 4]
7. Students understand how the data analysis methods work theoretically. [KBB5]
8. Students are able to critically review data analysis applications in which the methods taught methods have

been used. [KAR 4]

Course Content

In the analysis of biochemical systems, the tendency is to measure more and more of just a few samples, leading to complex multivariate data sets. Multivariate data analysis methods are often used to explore such sets.

This course covers a broad range of multivariate data analysis methods, for data preprocessing (e.g. variance stabilization), for exploration analysis (PCA, Clustering, nonlinear mapping), supervised analysis (ASCA, Classification, Random Forest). Furthermore, validation techniques such as cross validation and permutation testing are introduced and the interpretation of selected features in terms of function and networks is discussed. The course starts with an introduction on the properties of the different types of functional genomics data.

The main goal of this course is to teach students the theoretical properties of the data analysis methods and how to interpret the results of the multivariate methods and how this relates to the biological problem that is studied.

Additional Information Teaching Methods

- 10 lectures (2h each)
- 10 laptop seminars (4h each)
- Self-study for each of the 10 topics

Method of Assessment

There is a data analysis project in which students apply several methods to a dataset after which they have to interpret the results with respect to the statistical properties and the biological properties of the results. There is a theoretical test in which open questions are asked on the data analysis methods, their properties and especially on how to interpret their results. For the final grade, the weights are 1/3 and 2/3 for the project and the theoretical test, respectively. Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Entry Requirements

Bachelor in any science discipline, with an interest in biological data analysis. Basic level of linear algebra required. When in doubt please contact the coordinator.

Literature

Literature:

- Will be provided through Canvas.

Software:

- R.

Additional Information Target Audience

Admission to the course will depend on capacity, total number of applications, date of registration and background of the individual student. If the number of applications exceeds the capacity of the course, students may have to be selected and priority will be given in the following order:

- First-year students of the master Bioinformatics and Systems Biology (JD UvA and VU).
- Second-year students of the master Bioinformatics and Systems Biology (JD UvA and VU).
- Students of the master Computational Science (JD UvA and VU).
- Students of other master programmes.

Recommended background knowledge

Linear Algebra, Statistics with R

Cell Structures and Functions

Course Code	AM_470615
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English

Faculty	Faculty of Science
Course Coordinator	dr. S. Luirink
Examiner	dr. S. Luirink
Teaching Staff	dr. J.P. van Ulsen, dr. S. Luirink
Teaching method(s)	Lecture

Course Objective

In this course we will refresh and deepen the knowledge of the students in a number a selected topics of Molecular Cell Biology. Importantly, the objective is not to obtain a comprehensive overview of Cell Biology. After a general "refreshment of basic knowledge" and technical part, a few selected topics will be discussed in depth with guest lecturers (group leaders from different universities) based on reviews and primary research papers in their field of interest. An important aim is to learn how to read scientific papers and how to discuss scientific problems and controversies.

The selected topics circle around folding, modification and trafficking of proteins in cells (both prokaryotic and eukaryotic). How do proteins fold into their correct and active three-dimensional structure? How do proteins reach their correct destination in the cell (organelles, plasma membrane)? How do proteins insert and assemble in bacterial and mammalian membranes, how are they translocated across membranes? How are these processes related to disease and drug development? Along the way, the students will be informed on state-of-the-art methodsand techniques used in this field such as fractionation of cells, determination of protein structure, proteomics, molecular interaction/crosslinking techniques and visualization of cells and cell components by advanced microscopy.

Final attainment levels:

- Ability to interpret and discuss newly acquired primary literature in Cell Biology
- General knowledge of indicated cell biology topics and methodology relevant for the course at the level of the book "Molecular Cell Biology" (Alberts et al.)

Course Content

See "course objective".

Additional Information Teaching Methods

In the first part of the course (week 1 and part of week 2) students will study and discuss individually or in working groups part III (chapter 8,9; methods) and part IV (chapter 10, 12,13; internal organization of the cell) of the book: "Molecular Biology of the Cell" (Alberts et al.). Chapter by chapter the students will be guided through the book and questions will be answered in plenary sessions. Excursions will be organized to labs specialized in specific techniques (10 contact hours).

In the second part of the course (week 2-7; 20 contact hours) specific topics (protein trafficking, membrane protein function, importance of protein trafficking for Alzheimer's disease and tuberculosis) will be studied and discussed with lecturers from the VU and other universities. Each lecturer will present a seminar and discuss with the students very recent research papers and developments in his/her particular area of interest. The reviews and papers will be available via Canvas before the lectures.

The third part (last week) of the course will be used to study and to prepare for the exam.

Attendance at contact hours is mandatory.

This course follows the VU guidelines on the use of generative AI in education.

Method of Assessment

A written exam with open questions in which the Alberts book and print-outs of papers discussed by the lecturers can be used (open book exam). There will be appr. 4 open questions about the first part of the course covered by the indicated Alberts chapters and the technology lectures/excursions given in the first week. For the second part of the exam, the students have to study a primary paper that will be revealed before the study period starts (end of week 7). At the exam, appr. 5 open questions will be about this specific paper.

Entry Requirements

A bachelor degree in biology, medical biology, biomedical sciences or biochemistry.

Literature

Chapters from Molecular Biology of the Cell (Alberts, Johnson, Lewis, Raff, Roberts, Walter) Sixth or Fifth edition, Garland Sciences. Research papers presented by the lecturers: links available via Canvas 1 month prior to the course.

Additional Information Target Audience

Entry is restricted to BioMolecular Master students.

Additional Information

E-mail: s.luirink@vu.nl

Guest lecturers (subject to reservation): Dr. P. van der Sluijs (UU), Prof. L. Hamoen (UvA, Amsterdam), Prof. T. Sixma (NKI), Prof. E. Reits, Prof.. G. Jimenez (UAMC), Dr. E. Houben, Dr. C. Bunduc and Dr. P. van Ulsen (VU)

Recommended background knowledge

Basic (bachelor) knowledge of cell biology, microbiology, molecular biology and molecular genetics is required.

Drug Target Biochemistry and Signaling

Course Code	XM_0111
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. M.J. Smit
Examiner	prof. dr. M.J. Smit
Teaching Staff	prof. dr. M.J. Smit, dr. M.H. Siderius
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

Deepening of the knowledge of the biochemical aspects of drug targets and their role in cellular signaling in health and disease, with the aim to understand their importance in modern pharmacology.

Course Content

This course is designed for students with a background in the field of pharmaceutical and life sciences (or have followed Principles of Pharmaceutical Sciences) and an interest in the biological and pharmacological aspects of drug discovery. The course will focus on biochemical properties, including structures, of the main families of drug targets and their involvement in cellular signaling networks in health and disease (e.g. CNS-related disorders, cancer). Different strategies to identify, validate and modulate drug targets (e.g. ‘omics’ approaches, siRNA, CRISPR/Cas9, biologicals) will be addressed. New concepts in the architecture of signaling networks and mechanisms of signal transfer will be discussed.

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Written examination and assignments

Literature

Primary literature and handouts.
'Cell signaling', second edition. Authors: Wendell Lim and Bruce Mayer ISBN: 978-0-367-27937-0

Additional Information Target Audience

1DDS, 1BMS, MSc Bioinformatics and Systems Biology with Life Science background/BSc

Recommended background knowledge

BSc Pharmaceutical Sciences or equivalent or Principles of Pharmaceutical Sciences or BSc in Life Sciences

From genome to function

Course Code	AM_1290
Credits	6
Period	P2
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.R. Haanstra
Examiner	dr. J.R. Haanstra
Teaching Staff	dr. J.R. Haanstra, Y. Mulla, dr. S. Moco, drs. C.P. Kuijl
Teaching method(s)	Study Group, Written partial exam, Lecture

Course Objective

Students should be able to

- **Have insight** in the level of their foundational knowledge
- **Apply** the foundational knowledge in the cascade from genome to function on (novel) research data
- **Apply** the foundational knowledge in the cascade from genome to function on a scientific question within a group of peers
- **Work** effectively **together** in a group on an assignment
- **Discuss** the product of the group assignment

Course Content

The diverse functions of cells are executed by proteins. But these proteins need to be expressed from the genome in a correct and timely manner. So how do we get from the information in the genome to functional proteins? This question that is even more complex in eukaryotes where DNA modifications, post-transcriptional processes and posttranslational modifications will all work together to determine which proteins are present at a given time and whether they are active. And then a further layer of complexity comes from cellular localisation: many proteins function within a specific organelle. How do the proteins get in the organelle? And what is the function of compartmentalisation of (parts of) cellular pathways?

We will cover these topics with a special (but not exclusive) focus on the mitochondrion - an organelle that was probably originally a bacterium, but is now an important part of many eukaryotic cells.

Additional Information Teaching Methods

The course consist of interactive lectures on the basis of research papers and group work on in-depth questions.

The set-up of the course:

The course will start with a formative exam so that you can test your current level of knowledge.

Then we will have 3 weeks of lectures to refine your knowledge on topics important to this course. In these weeks you will start preparing for the group assignment in the second part of the course. After the lectures there will be a summative exam that will consist of 60% of your final grade.

In the second part of the course you will work on your group assignment with regular discussion/feedback moments. The output of this assignment will be a 3-page report. On the symposium day a random student from each group will be assigned to present the work of the group. The rest of the group will lead the discussion. The report , the groupwork and the discussion after the presentation will be assessed and will together consist of 40% of your final grade.

Method of Assessment

After the lectures there will be a summative exam that will consist of 60% of your final grade.

The output of this assignment will be a 3-page report. On the symposium day a random student from each group will be assigned to present the work of the group. The rest of the group will lead the discussion. The report , the

groupwork and the discussion after the presentation will be assessed and will together consist of 40% of your final grade.

Each of the two components (summative exam and assignment) need to be sufficient to pass the course

Literature

Research papers and powerpoint slides

Additional Information Target Audience

A solid background molecular genetics and biochemistry (from your bachelor) is essential to pass this course

Additional Information

You are not allowed to use generative AI in this course

Recommended background knowledge

Bsc level knowledge on genetics and biochemistry

Fundamentals of Bioinformatics

Course Code	X_405052
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. J.M. Urtasun Elizari
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	dr. H. Mouhib, dr. J.M. Urtasun Elizari, dr. ir. K.A. Feenstra, dr. D. Molenaar
Teaching method(s)	Seminar, Computer lab, Written partial exam, Lecture

Course Objective

Are you interested in Bioinformatics? Do you want to find out how biology can make an exciting application domain? Or do you want to learn what more you could do with your data, and with less effort? Enter here to start!

Fundamentals of Bioinformatics (FoB) is the starting course of the Bioinformatics master. It aims to give you a broad overview of important topics relevant to the field, with a focus on current open problems. You will be made aware of these open problems during practical sessions that will cover active research in a major Bioinformatics problem (mutation impact prediction). Based on your background, you will be assigned to separate classes where you will be working to fill gaps in your background knowledge in programming and/or biology.

After completing the course successfully:

- You can explain the fundamental principles of cell biology and the genetic information flow
- You can write short code to solve simple problems and complete skeleton scripts for biological applications
- You can read algebra and calculus equations
- You can become aware of gaps in your own background knowledge.
- You can address gaps in your background knowledge (programming, mathematics and molecular biology) and skills
- You are aware of the major issues when developing and benchmarking new methodology in bioinformatics research
- You can present and communicate a joined group projects carried out with students of diverse backgrounds (Interdisciplinarity) in a respectful way
- You can apply and understand fundamental methods in bioinformatics, e.g. sequence alignment, impact prediction, and to understand and use fundamental concepts in bioinformatics, e.g. homology, evolutionary conservation, and gene function

Fundamental knowledge:

Students will have the opportunity to level up their knowledge of molecular biology, maths (linear algebra and calculus) and programming (Python), according to their needs.

Course Content

Topics:

- Evolution, genomics, sequence alignment, Blast/PSI-Blast, FAIR data, benchmarking, machine learning, genome assembly, mutation impact prediction, protein stucture analysis, substitution matrices, scientific writing

Project:

- Group project to benchmark different mutation impact prediction approaches. The groups will be composed to include different BSc backgrounds.

Background knowledge (1 out of 3, see below):

- Molecular cell biology
- Programming and computational thinking in Python
- Calculus and linear algebra

Additional Information Teaching Methods

- 8 lectures (two-hour lecture in the morning, two days per week) Exercises during/in between lectures
- 12 computer practicals for the group project - compulsory attendance (online or on campus) (two hour sessions following the morning lectures, two days per week), supervised.
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Conversion classes for background knowledge:

- Molecular cell biology (7 x double lectures)
- Programming and computational thinking in Python (10 computer practicals)

Depending on your background, you will be assigned 1 of the 2 conversion classes. Students that are already well versed in both topics (molecular biology and programming) may be assigned to the following class, which is organised as part of the Introduction to Systems Biology:

- Calculus and linear algebra (10 maths classes)

Method of Assessment

- Project work [~40%] :
- Project report and/or presentation
- Digital exam to assess [~60%]:
- Lecture topics
- Concept learned during the project
- Concepts from conversion classes

(assessment of practical assignments may be adjusted based on ongoing developments in generative language models)

Further assessment and grading details will be posted on Canvas (resits and compensation rules)

Entry Requirements

Bachelor in any science discipline (including medicine) or strong programming background, and an interest in biological problems.

Literature

- course material (slides, scientific papers) on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum Understanding Bioinformatics Garland Science 2008 ISBN-10: 0-8153-4024-9

Additional Information Target Audience

M Artificial Intelligence
M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Oncology
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will help you to set this up.

Recommended background knowledge

Programming in Python, Molecular Biology, some basics of calculus and linear algebra

Internship M BMOL I

Course Code	AM_1294
Credits	30
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I. van der Veen
Examiner	dr. I. van der Veen
Teaching Staff	dr. S.A.K. Jongkees, dr. V. Kokkoris
Teaching method(s)	

Course Objective

The aim of a master internship is to train students to conduct scientific research in a typical “on the job” situation.

The student:

- is able to independently find scientific information and knows how to analyse, summarize and critically evaluate this for the benefit of his or her own research question;
- has the ability to use the principles from different disciplines (such as Biochemistry, Molecular Cell Biology, Biophysics) in the design of research plans, the execution of research, and the analysis of the results;
- has command of the relevant research techniques and laboratory procedures, including safety procedures, and the ability to solve emerging problems;
- has command of the use of computer software relevant for the field;
- is able to document and communicate experimental results in a labjournal, written report and oral presentation;
- can analyse and evaluate planning, execution and results of research independently and critically;
- is able to collaborate with researchers of various disciplines;
- can contribute to scientific discussions about plans, results and consequences of research;
- can reflect on ethical aspects of research and applications of research;
- can evaluate his or her own functioning in the internship, both by reflection and in discussions with others;
- has obtained a good impression of a potential future field of career.

Course Content

All information about the internship can be found in the internship manual, which is available from the CANVAS ‘Biomolecular Sciences’ community site.

Additional Information Teaching Methods

Internship

Method of Assessment

Grades will be given for attitude, execution, report and presentation

Entry Requirements

Students may start the internship only if they have acquired 12EC from period 1 and 2

Additional Information

This course follows the VU guidelines on the use of generative AI in education.

Internship M BMOL II

Course Code	AM_1306
Credits	33
Period	Ac. Year (September)
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. I. van der Veen
Examiner	dr. I. van der Veen
Teaching Staff	
Teaching method(s)	

Course Objective

The aim of a master internship is to train students to conduct scientific research in a typical “on the job” situation.

The student:

- is able to independently find scientific information and knows how to analyse, summarize and critically evaluate this for the benefit of his or her own research question;
- has the ability to use the principles from different disciplines (such as Biochemistry, Molecular Cell Biology, Biophysics) in the design of research plans, the execution of research, and the analysis of the results;
- has command of the relevant research techniques and laboratory procedures, including safety procedures, and the ability to solve emerging problems;
- has command of the use of computer software relevant for the field;
- is able to document and communicate experimental results in a labjournal, written report and oral presentation;
- can analyse and evaluate planning, execution and results of research independently and critically;
- is able to collaborate with researchers of various disciplines;
- can contribute to scientific discussions about plans, results and consequences of research;
- can reflect on ethical aspects of research and applications of research;
- can evaluate his or her own functioning in the internship, both by reflection and in discussions with others;
- has obtained a good impression of a potential future field of career.

Course Content

All information about the internship can be found in the internship manual, which is available from the CANVAS ‘Biomolecular Sciences’ community site.

Additional Information Teaching Methods

Internship

Method of Assessment

Grades will be given for attitude, execution, report and presentation

Entry Requirements

Students may start the internship if they completed the first internship

Additional Information

This course follows the VU guidelines on the use of generative AI in education.

Introduction to Systems Biology

Course Code	X_428565
Credits	6

Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. G. Stolf Jeuken
Examiner	dr. D. Molenaar
Teaching Staff	dr. M.T.M. Wortel, dr. D. Molenaar, prof. dr. F.J. Bruggeman, dr. G. Stolf Jeuken
Teaching method(s)	Seminar, Study Group, Written partial exam, Lecture

Course Objective

Introduction to Systems Biology is the starting course of the Bioinformatics and Systems Biology master (together with Fundamentals of Bioinformatics). To emphasise the importance of systems thinking biology, this course introduces the principles of systems biology, with a primary focus on the analysis of metabolic networks using stoichiometric modelling.

The course explores how genomic information can be translated into computational models to understand and predict cellular behaviour, linking genotype and phenotype. The course will see the application of these models in biotechnology and medicine, utilizing mathematical concepts (particularly linear algebra) and computational tools.

After completing the course successfully [Bloom taxonomy]:

- You understand the challenge of linking genomic information (genotype) to cellular behaviour (phenotype). [understand]
- You can write short code to solve simple problems and complete skeleton scripts for computational modelling of biological processes [apply]
- You can solve mathematical problems using algebra. [apply]
- You are able to represent biological information, specifically metabolic networks, in a computable format using mathematical concepts. [evaluate]
- You are able to use computational tools and algorithms to analyse metabolic networks. [apply]
- You appreciate the importance of experimental data and design in systems biology. [analyse]

This will enable you to understand and perform computational analysis related to:

- Systems Biology Research: Using metabolic network models to understand cellular growth under different conditions.
- Biotechnological Applications: Identifying genes to modify in microbial genomes for the production of valuable products like biofuels.
- Medical Applications: Identifying potential drug targets within the metabolism of pathogens or cancer cells for therapeutic intervention.

Fundamental knowledge:

Students will have the opportunity to level up their knowledge of molecular biology, maths (linear algebra and calculus) and programming (Python), according to their needs.

Course Content

The course will cover the following key topics:

- Genome Annotation and Network Reconstruction: Decoding genomic information to build metabolic networks.
- Stoichiometric Principles: Understanding chemical element and charge conservation in biochemical reactions, stoichiometric coefficients, and balancing reactions.
- Reaction Networks and Stoichiometric Matrix: Representing networks mathematically using the stoichiometric matrix and formulating mass balance equations.
- Steady-State Analysis: Concepts of steady-state, null space of the stoichiometric matrix, steady-state flux distributions, and the rank theorem.
- Linear Programming and Flux Balance Analysis (FBA): Introduction to FBA as a core computational method for analysing metabolism.

Conversion classes:

Depending on their needs, students can choose to attend conversion classes where they will be working to fill gaps in their knowledge in mathematics (linear algebra and differential equations), programming or biology.

Additional Information Teaching Methods

- Lectures
- Workgroups
- Conversion classes in biology or mathematics or programming

Method of Assessment

A written exam. To pass the exam, the grade must be 5.5 or higher. Retakes of the exam take place once a year. The last obtained pass grade will count as the final grade for the course.

Entry Requirements

Bachelor in any science discipline (including medicine), with an interest in applying mathematical techniques in biology. When in doubt please contact the coordinator.

Literature

A course syllabus will be provided

Additional Information Target Audience

Students with a bachelor degree in Biology, Bioinformatics, Mathematics, Physics or similar background

Recommended background knowledge

Mathematics at high-school level

Lifecycle of Research Projects

Course Code	AM_1293
Credits	6
Period	P1, P1
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. E.N.G. Houben
Examiner	dr. E.N.G. Houben
Teaching Staff	dr. E.N.G. Houben
Teaching method(s)	Seminar

Course Objective

Course objectives

Knowledge goals:

Designing an effective research question

1. To be able to develop a clear, innovative, impactful, and researchable research question.
2. To be able to demonstrate the potential for significant scientific, societal, and/or economic impact within your research question or topic.
3. To understand which types of research questions are likely to be funded and the reasons behind their selection.

Understanding the main principles of successful scientific project proposals

1. To thoroughly understand the essential elements that make up a good scientific research proposal.
2. To understand how to design projects that push the boundaries of scientific knowledge, while ensuring that the projects remain practically feasible.
3. To be able to articulate the potential impact and relevance of the research, while explaining how the project can contribute to the field scientifically, socially, and/or economically.

Understanding the basics of scientific project management

1. To understand basic project management concepts and frameworks applicable to scientific research.
2. To be able to develop project plans that include clear objectives, milestones, and deliverables aligned with scientific goals.

3. To be able to identify potential risks in research projects and strategies for mitigating these risks.
4. To understand techniques for monitoring project progress against the work plan, and making adjustments as necessary to stay on track.

Understanding the differences between scientific projects in academia and industry

1. To understand the foundational differences in objectives, culture, and operational frameworks between academic and industrial scientific settings.
2. To understand how the goals of scientific projects differ in academia and industry, focusing on basic versus applied research, and how these impact project design and expectations.
3. To be able to analyze how expectations for project outcomes and deliverables differ, particularly in terms of publication, patenting, and commercialization.
4. To be able to compare the level of innovation and risk tolerance in academic research versus industry projects, including how failure is perceived and managed.
5. To understand about the nature of team collaborations in both settings, including interdisciplinary teamwork in academia and cross-functional teams in industry.

Skill-based goals:

1. To be able to formulate comprehensive research questions that encompass the essential elements of a well-structured and fundable research proposal. These elements include innovativeness, feasibility, relevance to existing and future research, and the potential for societal impact
2. To be able to write a research proposal, mirroring the structure and components typical of proposals submitted for NWO or European grants.
3. To be able to write a basic project management plan that should supplement the research proposal.
4. To be able to pitch a research project and answering scientific and project feasibility questions.
5. To be able to work in a team through group assignments.
6. To be able to provide constructive feedback on research proposals of peers.

Attitudinal goals:

To be able to shift from a student’s perspective to that of a young professional engaged in scientific projects. This transformation will be facilitated through an exploration of key concepts such as project ownership, project management, the long-term objectives of the project, and its feasibility.

Professional and career goals:

To be able to balance innovativeness with feasibility by adopting a broader perspective on scientific projects, and acquiring project management techniques tailored for life science research. This preparation will better equip students for their initial professional roles in either academia or industry.

Course Content

The course will cover a broad range of skills and knowledge that young professionals need to possess to successfully run and be engaged in research projects.

It will guide students through the process of formulating impactful research questions that align with funding priorities, crafting detailed proposals, and balancing innovation with feasibility in project design.

Additionally, the course will cover the unique aspects of conducting research in both academic and industry settings, ensuring students understand how to navigate the distinct dynamics of each environment throughout the research project's lifecycle.

Moreover, the course seeks to shift students' perspectives, encouraging them to think and act as young professionals and prepare them for successful careers in diverse research environments.

Additional Information Teaching Methods

The course activities consist of:

1. Series of lectures
2. Group assignments: writing research proposal and project management plan, preparing a presentation
3. Individual assignments: presenting, participating in discussions, peer-reviewing

Method of Assessment

The final course grade is based on both group and individual assignments. Group assignments include writing a project proposal and a management plan (40%) as well as preparing a presentation (10%). Individual assignments consist of presenting and participating in discussions (20%), peer-reviewing (20%), and contributing to the group assignment (10%). A 5.5 or higher must be obtained on all assignments to pass the course.

Entry Requirements

The course is mandatory for the 2nd year Biomolecular MSc students and is most-suited for students who finalized their 1st internship and therefore already have some experience and insights about how scientific research projects are organized.

Literature

Course material will be provided on CANVAS.

Additional Information

Please note that a general VU policy on the use of AI tools applies to all assignments. To find out more - please follow this link

Explanation Canvas

All communication about the course will be through Canvas.

Molecular and Cellular Toxicology

Course Code	XM_0113
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. J.C. Vos
Examiner	dr. ir. J.C. Vos
Teaching Staff	dr. ir. J.C. Vos, dr. S. Moco, dr. A. Wilmes, prof. dr. P. Jennings
Teaching method(s)	Lecture

Course Objective

Students will gain knowledge on various ways cells can be compromised by exposure to exogenous chemicals, how they defend themselves against specific perturbations and the methodological aspects in determining and quantifying these processes.

Course Content

Exposure of cells to chemicals (including drugs and drug metabolites) can cause perturbations compromising the cells' ability to function which may lead to cell death and tissue injury eventually compromising health. The course will introduce and expand on such concepts including cell injury processes, cell death, cellular defence mechanisms (e.g. p53 pathway, Nrf2 pathway, unfolded protein responses), mitochondrial toxicity and immunotoxicity.

The following aspects will be explored:

- types of adverse drug reactions,
- role of xenobiotic biotransformation (metabolism) transport in adverse drug reactions and nuclear receptors,
- reversible and irreversible interactions of toxic drugs with biological macromolecules,
- stress response pathways - cellular adaptation to exposure to chemicals and reactive oxygen species;
- cellular and molecular mechanisms leading to toxic effects,
- mitochondrial toxicity,
- immune-mediated toxicity,
- cellular in vitro models used in toxicity studies,
- cellular assays used to detect stress-responses and mitochondrial dysfunction including transcriptomics, metabolomics and proteomics.

Additional Information Teaching Methods

Lectures and self-study.

Method of Assessment

Written exam

Literature

Literature will be provided.

Additional Information Target Audience

mDDS, mBMS, mCHEM

Recommended background knowledge

Bachelor Pharmaceutical Sciences, Chemistry, Biomedical Sciences , Medical Natural, Sciences, Medical Biology or equivalent.

Molecular Infection Biology

Course Code	AM_470657
Credits	6
Period	P2
Course Level	600
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. W. Bitter
Examiner	prof. dr. W. Bitter
Teaching Staff	prof. dr. W. Bitter, A. Speer, drs. C.P. Kuijl
Teaching method(s)	Seminar, Practical, Study Group, Written partial exam, Lecture

Course Objective

- To understand how the pathogen interacts with its host and how this is studied: in vitro studies, (alternative) in vivo models, different approaches of mutant screening
- To understand the genetic variation of microbial pathogens (viruses and bacteria) and the effect this variation has on host adaptation.
- To understand what virulence factors are and how they are identified and regulated by the pathogen.
- To apply the acquired knowledge to interpret scientific literature and scientific hypotheses regarding host-pathogen interactions.

Course Content

The rapid expansion of genomic, proteomic, and RNA sequencing data from both microbial and eukaryotic hosts, combined with continuous advances in molecular biology, has enabled detailed analysis of the molecular interactions between pathogens and their hosts. This knowledge is essential, as we are constantly confronted with emerging infectious agents and the re-emergence of historical diseases, necessitating the development of new vaccines and antimicrobial therapies.

A central challenge lies in selecting the most appropriate techniques for addressing specific research questions and interpreting potentially conflicting results arising from different experimental approaches, i.e. the choice of the technique or approach used already determines which factors you can and can not identify .

This course is designed to provide a comprehensive understanding of molecular biology as it pertains to infectious agents. It focuses on the application of molecular biology to investigate host-pathogen interactions of pathogenic bacteria and viruses—with an emphasis on virulence factors as well as the genetic susceptibility of hosts to infection. The course aims to equip students with the specialized knowledge and analytical skills required to critically evaluate primary literature in the field of molecular medical microbiology.

Additional Information Teaching Methods

The course has three different parts: lectures, literature presentations and rebuttal assignment. In the literature assignment students will use their acquired knowledge from the lectures to dig deeper into specific aspects of host-pathogen interactions based on a recent high-profile publication. In the rebuttal assignment the students will work on an actual scientific publication that has come back from the reviewers to interpret the comments and use these to write a rebuttal using new data.

Contact hours:

Lectures: 20h
Literature Workshop: 16-20h (depending on number of students)
Rebuttal assignment: 5 h

Method of Assessment

- written exam (50% of final mark and should be minimally 5.5)
- literature discussion (workshop, 25% of final mark)
- practicals/assignment (25% of final mark)

Entry Requirements

Bachelor's course 'Infectious Diseases' and 'Immunology' or an equivalent course in Microbiology and Molecular Biology.

Literature

List of important articles that are supporting the lectures will be provided on CANVAS

Additional Information Target Audience

Students with a keen interest to study the interaction between a pathogen and its host, from a practical as well as a theoretical point of view. Compulsory course for specialization Infectious Diseases.

Additional Information

Guest lectures:
Dr. Lia van der Hoek, AUMC Amsterdam, identification of novel viral pathogens
Dr. Wiep Klaas Smits, LUMC Leiden, working with gut pathogens

Custom Course Registration

You need to register yourself for the course. The course coordinator will register you for the practicals/assignments.

Oncogenesis

Course Code	M_OONC03
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Medicine
Course Coordinator	prof. dr. R.H. Brakenhoff
Examiner	prof. dr. R.H. Brakenhoff
Teaching Staff	S. Biemans, dr. S.B. Baglio
Teaching method(s)	Seminar, Lecture

Course Objective

The aim of this course is to acquire state-of-the art knowledge of and deeper insights into the general concepts of oncogenesis and basic tumor biology. These include (but are not limited to) the different genetic alterations and classes of genes involved in tumor formation, their role in cellular signaling, DNA repair and cell cycle control, the (molecular) tools to study carcinogenesis including NextGen sequencing, CRISPR-Cas9 gene editing, model systems such as transgenic mouse models and high-throughput functional genomics.

In addition, hereditary cancer predisposition syndromes, exogenous factors involved in oncogenesis with emphasis on viruses will be studied. Finally, we will review markers of disease progression, genetic counselling, screening, as well as intervention options.

The students will be given an up-to-date overview of the current knowledge and opinions in this exciting field. Furthermore, the students will acquire deep insight into how to interpret literature on specific topics in oncogenesis and present their findings in a clear and concise manner.

Course Content

The course on Oncogenesis consists of three parts. The first part involves morning sessions with lectures given by experts on various topics of oncogenesis. The second part involves a literature (self-) study on a specific familial cancer syndrome followed by a group-presentation in the last week of the course. The third part involves discussion on one of the “hallmarks of cancer” for which students will summarize lectures with the aid of the handouts that will become available on CANVAS and additional literature. The latter two parts will be executed by groups of 4 students in a way that all students will be involved in a presentation.

Topics include:

1. Latest updates on the roles of cancer genes and cancer pathways
2. Modern cancer genomics and (epi)genetics including Next Gen sequencing
3. Mechanisms, models and molecular methods, including CRISPR-Cas9 genome editing
4. Viruses causing cancer
5. Clinical aspects of cancer and precancer biology
6. Cancer predisposition syndromes

Additional Information Teaching Methods

± 45 interactive lectures given by experts during morning sessions
± 8-10 hallmark discussions given by student groups
± 8-10 interactive literature presentations given by student groups

The course has approximately 56 contact hours with teachers and course coordinators.

Method of Assessment

The course will be concluded with a 2.5 hours examination (open questions) on the contents of the lectures. Handouts of the presentations, including those of the literature studies, will be available on CANVAS. There will be an examination training during the course. The hallmark discussions and presentations of the literature studies will be assessed as well.

The final mark of this course will be determined by the average of the scores for the examination (counting for 70%), the hallmark discussion (counting for 10%), and the literature presentation (counting for 20%). Students need to be present and participate actively during the course. Absence is only allowed with reason and needs to be reported on beforehand.

The (review) articles posted on CANVAS and ‘the Biology of Cancer’ (Robert A. Weinberg, second edition) are recommended as a reference work for deeper background knowledge and to facilitate studying the different subjects.

Literature

The course reader (available on CANVAS) contains (review) articles on the various topics.

Additional Information Target Audience

Only for first year Master Oncology students.

Note: Students from other programs will need to be approved by the master Oncology Admissions Board to be admitted to the course.

Additional Information

The students will have acquired insight in the latest developments in the field of oncogenesis, cancer genetics and technologies to investigate the molecular aspects of cancer. Furthermore, the students will be able to integrate knowledge and insight in the formation of opinions concerning research questions about the development of cancer, and learn how to review and present scientific work.

Coordinators:

Prof. dr. R.H. Brakenhoff
Dr. R.M.F. Wolthuis
Dr. S.R. Baglio

Custom Course Registration

All Master Oncology students are required to attend the compulsory courses of the programme. Master Oncology students must register for all courses using the registration module. All activities for which you have registered will be displayed in your personal timetable, including any changes to the timetable.

If you have not registered for a course, you will not be admitted to that course, you will not be assigned to a group, you will not be able to use CANVAS, you will not have a timetable, your grades will not be recorded, etc. In short, you will not be able to participate. Register via the registration module.

This course is open to VU students and external students from other master programmes.

Prerequisites: background knowledge of oncology and immunology. Participation is only possible if the number of students in the course does not exceed 40 and after approval by the master Oncology admission board. To apply, you must send your CV, letter of motivation and transcript of records to the programme coordinator by email: masteroncology.vu@amsterdamumc.nl

Students will be notified of their acceptance at the beginning of July.

Principles of Drug Targets

Course Code	XM_0105
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. R. Leurs
Examiner	prof. dr. R. Leurs
Teaching Staff	prof. dr. R. Leurs
Teaching method(s)	Seminar, Lecture

Course Objective

To gain insight into general steps in the process of drug discovery & development, understanding of the molecular basis of target-drug interactions and the role of pharmaco- and toxicodynamics.

- You will get acquainted with the various processes essential for drug discovery & development
- You will understand the structure and function of different drug targets
- You can explain how enzymes, receptors and DNA can be targeted by drugs
- You will understand the molecular basis of target-drug interactions and how drug-target interactions can be quantified
- You will learn the importance of drug metabolism and safety assessment

Course Content

This course is designed for students with an interest in life sciences and the biotech/pharmaceutical industry but without prior education in this field. A general introduction will be given to the process of drug discovery, drug design and synthesis, drug development and drug safety assessment. Subsequently, potential drug targets, mechanisms of drug actions (including drug- receptor/enzyme interactions and dose- response relations), drug disposition (including pharmaco-/toxicokinetics) and drug toxicity will be discussed. Various modern developments and tools will be illustrated by recent applications in the field of drug discovery.

Additional Information Teaching Methods

Lectures, tutorials, self-study, assignments.

Method of Assessment

Written exam and assignment(s)

Literature

Patrick, G., An Introduction to Medicinal Chemistry 6th ed., Oxford: Oxford University Press. 20017, ISBN: 978-0-19-874969-1, mainly Part A (Drug targets) and Part B (PDPK). Next to this textbook, some primary literature will be used to illustrate certain topics.

Additional Information Target Audience

This introductory course is primarily designed for students mDDS**without** a Bsc degree in Pharmaceutical Sciences or SBI. Accessible for students from mCh, mBMS (Biological Chemistry track) or mSBI (life track) master.

The course is **not** accessible for people with a BSc degree in Pharmaceutical Sciences or SBI.

Recommended background knowledge

Knowledge of basic biochemistry and organic chemistry

Professionalism in Biomolecular Science year 1

Course Code	AM_1273
Credits	0
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V. Chubar
Examiner	V. Chubar
Teaching Staff	K. Wessels MSc, dr. J.I. Geertsema, V. Chubar
Teaching method(s)	Meeting, Study Group, Symposium, Participatory Teaching Method, Tutorial

Course Objective

The Professionalism in Biomolecular Science (PIBS) course intends to activate our students to thinking more professionally and in an engaged manner about their study track, internships and future careers as scientists.

Course Content

During year 1, we start preparing you for (finding) the internship, with a dedicated first workshop and internship symposium.

All workshop and assignments in year 1 aim to provide some context, preparation & feedback on the professional skills that scientific professionals are often required to engage in when they're not in the lab.

Workshop topics include communication & feedback skills, peer-review, presentation skills, poster and scientific figure design, self-reflective practice, networking, goal-setting and project planning.
We'll discuss these topics, analyze examples, and practice using both in-class activities and small assignments.

Students are also encouraged to think about their second-year study plans.

During year 2 (AM_1277) you will put what we've learned from the year 1 & 2 workshops and assignments into practice by putting together a learning portfolio with a professional development project.

As part of this course, our students are encouraged to get involved in organizing and attending our alumni career event, and a student research symposium, where student internship projects can be presented as scientific posters to peers and local academics.

Additional Information Teaching Methods

Year 1 & 2 active workshops, Year 2 feedback session, self-guided project work with mentors.

Method of Assessment

This course fully runs across 2 academic years. All components must be completed & submitted in order to obtain 3 EC study credits during year 2 (course code = AM_1277).

For year 1, attendance & active participation during workshops with submitted assignments.

For year 2, attendance & active participation during workshops with submitted assignments and submitted final learning portfolio.

Deadlines for course schedule will be communicated via the course manual & using CANVAS.

Additional Information Target Audience

Students of the master degree Biomolecular Sciences.
This course can and should be undertaken by students who have commenced their studies in 2019-2020 or later.

Additional Information

Please note that a general VU policy on the use of AI tools applies to all assignments. To find out more - please follow this link

Custom Course Registration

Workshops are given in small parallel groups of 10-15.
Canvas instructions will follow at the beginning of the academic year.
After the course has started, students should sign up for workshop groups using Canvas, to a group that matches their own study schedule.

Explanation Canvas

Deadlines for course schedule will be communicated via the course manual & using CANVAS.

Professionalism in Biomolecular Science year 2

Course Code	AM_1277
Credits	3
Period	Ac. Year (September)
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	V. Chubar
Examiner	V. Chubar
Teaching Staff	dr. J.I. Geertsema, V. Chubar, K. Wessels MSc
Teaching method(s)	Meeting, Study Group, Symposium, Participatory Teaching Method, Tutorial

Course Objective

The Professionalism in Biomolecular Science (PIBS) course intends to activate our students to thinking more professionally and in an engaged manner about their study track, internships and future careers as scientists.

Course Content

The students will now put what they learned from the year 1 and 2 workshops and assignments into practice by producing their own learning portfolio.

Topics covered during year 2 themed workshops include: Pitching an idea or project successfully, job market orientation, job application & CV preparation, open science & data management.

For the final learning portfolio, students need to justify, design, carry out, and later reflect on their own self-directed professional development project (major project).

We also organize a "project inspiration session" to prepare students for designing their project. The purpose of the project is to give students the opportunity to build their CV and fill niches in their own professional development and transferrable skill-building away from the lab.

Students will use input and feedback from their tutor, project mentors, peers, and other experts in the local scientific community to bring their professional development project (major project) ideas to fruition during the final semester of their degree.

As part of this course, our students are encouraged to attend the alumni career event and our student research symposium, where student internship projects can be presented as scientific posters to peers and local academics.

Additional Information Teaching Methods

Year 2: Project inspiration session in period 1, 2x workshops, self-guided project work with mentors and tutor, final portfolio assessment

Method of Assessment

Submitted homework assignments.
Students are obliged to attend workshops and participate actively.

A submitted final portfolio with major project.
This course will be assessed with a pass/fail.

Additional Information Target Audience

This course is compulsory for all Biomolecular sciences master students.

Additional Information

Deadlines for the course schedule will be communicated via the course manual & using CANVAS.

Please note that a general VU policy on the use of AI tools applies to all assignments. To find out more - please follow this link

Custom Course Registration

Workshops are given in small parallel groups of 10-15. Canvas instructions will follow at the beginning of the academic year. After the course has started, students can sign up for workshop groups on Canvas.

Explanation Canvas

Deadlines for the course schedule will be communicated via the course manual & using Canvas.

Protein Science

Course Code	AM_470145
Credits	6
Period	P1
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Bald
Examiner	dr. D. Bald
Teaching Staff	dr. D. Bald, dr. H. Mouhib, dr. ir. Y.J.M. Bollen, dr. D.P. Geerke
Teaching method(s)	Study Group, Lecture

Course Objective

The student:

1. knows and understands principles of protein structure, function, dynamics, regulation, inhibition, interaction and engineering
2. can explain and predict protein function based on protein structure and understanding the molecular properties of amino acid residues.
3. knows and understands the principle of current methods for protein investigation (e.g. structure determination, activity assays, overproduction, purification, interaction)
4. can analyze the strong and weak points of Protein Science techniques and can correlate an open question with a suitable technique/approach.
5. can analyze experiments in Protein Science and design new experiments.

Course Content

We will start with a repetition of protein structure and function.

Subsequently, we will focus on methods in protein science and also on more specialized properties of proteins important in fundamental research, biomedicine or biotechnology. Finally we will deal with case studies on selected proteins.

Planned lecture topics include:

- Protein Structure, Protein Function, Protein Dynamics, Control of
- Protein Function, Antibiotic action and antibiotic resistance,
- Protein over-production and purification, Protein structure and interaction, Protein Engineering.

Planned case studies:

- New antibiotics, Protein hydrophobicity and folding,
- Molecular modeling/docking, Protein-protein Interaction modulation.

Additional Information Teaching Methods

Lectures and work discussions, accompanied by self study to prepare for the lectures and to discuss the material presented in lectures.

Method of Assessment

Written exam (100%)

Entry Requirements

Participation in prior Bachelor-level courses on proteins/biochemistry is expected.

Literature

No specific book required. You can use any Biochemistry textbook, e.g. "Voet and Voet", for repetition.

A more concise alternative is "Protein Structure and Function" by Petsko/Ringe (somewhat old, but nicely written).

You will receive reading/study material, mainly reviews and original articles on relevant topics.

Examples of scientific literature: Jin et al, Nature 2020, Goojani et al. Biochim. Biophys. Acta 2020, Ple et al. Nature Communications 2022.

Additional Information Target Audience

The main target audience are Masters students Biomolecular Sciences. Additionally, students from Masters programs Biomedical Sciences, Bioinformatics, Pharmaceutical Sciences and Medical Natural Sciences can participate.

Additional Information

This course follows the VU guidelines on the use of generative AI in education (https://vu.nl/en/student/examinations/generative-ai-your-use-our-expectations)

Recommended background knowledge

Participation in prior Bachelor-level courses on proteins/biochemistry is expected.

Quantitative Single-cell Biology

Course Code	XM_0084
Credits	6
Period	P6
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. F.J. Bruggeman
Examiner	prof. dr. F.J. Bruggeman
Teaching Staff	dr. E. Tutucci, prof. dr. F.J. Bruggeman
Teaching method(s)	Seminar, Computer lab, Practical, Lecture

Course Objective

Motivation - Modern eukaryotic cell biology, microbiology and molecular systems biology, and their applications in medicine, rely heavily on single-cell approaches. Live-cell imaging with fluorescence microscopy, flow cytometry and single-cell sequencing are becoming standard tools in labs. With it come theoretical challenges too. Methods from statistical biophysics, quantitative image analysis and machine learning are being used to predict, analyze raw data, and understand experimental outcomes. Thus, researchers need to be trained in the experimental and theoretical aspects of quantitative single-cell biology.

After successful completion of this course:

- You can give an overview of the most modern single-cell analysis theoretical and practical method
- You can explain the key facets of quantitative single-cell analysis
- You can explain the basic theory and simulation methods in addition to experimental single-cell analysis methods, which include single-molecule RNA imaging (smFISH), fluorescent-tagging of proteins, fluorescence microscopy and flow cytometry
- You can use and adapt custom-written Python code to analyze single cell data
- You can create new analysis approaches for specific research questions and can use and write scripting of analysis pipelines
- You can analyze experimental data (imaging analysis of fluorescence microscopy data, flow cytometry and growth measurements)
- You can investigate how single-cell microorganisms adapt to changing environments
- You can explain concepts including fitness, gene expression noise, adaptation of metabolism and population heterogeneity
- You can use computational methods to analyze the data (i.e. quantify single-cell RNA and protein expression changes in space and time), which will be compared to expectations deriving from theory and stochastic simulations
- You can report the result of their analysis in a research paper-like format to demonstrate your ability to convey complex biological question and approaches
- You are familiar with the basic concepts, successes, and challenges of the current field of quantitative single-cell biology
- You are able to analyze complex biological questions

Course Content

The main topics covered by the course are as follows:

- Key single-cell phenomena from cell birth to division, single-cell stochasticity;
- Relations between microbial fitness, dynamic environments, phenotypic heterogeneity and phenotypic diversification;
- Modern Single-cell techniques, e.g. microscopy (single molecule RNA imaging and fluorescent protein tagging) and flow cytometry;
- Data analysis: stochastic simulation, image analysis, cell-segmentation, cell-tracking and statistical methods using a combination of Mathematica, Phython and ImageJ;
- Experimental methods: fluorescent protein labeling, single-molecule RNA imaging methods, real-time imaging of cell growth and fluorescent reporters; flow cytometry.

Additional Information Teaching Methods

Teaching Methods

- Lectures (Week 1: 14 hours; Week 2: 8 hours; Week 3: 6 hours)
- Work groups, including data analysis, report writing and Q&A sessions (Week 1: 12 hours; Week 2: 21 hours; Week 3: 21 hours; Week 4: 7.5 hours)
- Lab visit (1 afternoon)

Method of Assessment

Theoretical exam (70%) and project (30%).

Further assessment and grading details will be posted on Canvas (resits and compensation rules).

Entry Requirements

Basic cell biology, basic statistics and basic computer skills.

Literature

Course syllabus and PowerPoints provided during the course.

Additional Information Target Audience

MSc Bioinformatics and Systems Biology
MSc Biomolecular sciences

Additional Information

Students are required to bring a laptop

Replacement, Refinement and Reduction of animal experiments applying human cells in culture

Course Code	XM_0148
Credits	6
Period	P5
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. A. Wilmes
Examiner	dr. A. Wilmes
Teaching Staff	dr. ir. J.C. Vos, dr. A. Wilmes, prof. dr. P. Jennings
Teaching method(s)	Seminar, Study Group, Other, Lecture

Course Objective

To get familiar with the principles of the 3R and cell cultures techniques including for the application in pharmacology and toxicology.

Course Content

The utilisation of whole animal studies in biological applications such as drug development has inherent ethical concerns, thus there is an impetus to develop non-animal models which may even bring advantages. Human cell culture offers a great tool in various research areas, including in the field of pharmacology and toxicology. Human cell culture systems often provide numerous advantages over animal experiments, including human relevance, reduced cost and the possibility to study mechanistic approaches. Nevertheless, choosing the best available cell model for the respective research question is critical. Furthermore, recent advances and cell culture methods and new techniques, including 3D cultures and organ on chip models evolve rapidly in this field. This course covers the basic understandings in cell culture methods as well as recent developments of advanced culturing techniques and compares different available human cell models, including (cancer) cell lines, primary cells or stem cells, such as induced pluripotent stem cells (iPSC).

In this course, the students will be familiarized with:

- The use of animals by humans: from domestication to the laboratory
- Ethical issues related to human and animal use for drug safety assessment.
- Russell and Burch and the origins of the 3Rs (Replacement, Reduction, Refinement). The principles of the humane experiment.
- Different cell culture media and coatings, including serum free media
- Physiological conditions for cell culture
- Cell proliferation, cell differentiation and respective laboratory techniques (Cell counting, cell viability assays, subculturing, cryopreservation, cell differentiation protocols)
- Primary cell culture and culture of human cell lines, including senescence, immortalization and transformation
- induced pluripotent stem cells (iPSC) culture and differentiation into target cells
- Advances cell culture techniques, including co-culture, 3D culture, organoids and microphysiological systems (MPS)/ organ on a chip
- Sterile handling and sources of contaminations, including mycoplasma
- Cell quality assurance techniques, including authentication, characterisation and validation

Additional Information Teaching Methods

Lectures and assignments

Method of Assessment

Exam and assignments.

Literature

Recommended books:

1. Culture of animal cells. R. Ian Freshney. Eight Edition (March 2021). Wiley-Blackwell. ISBN: 978-1-119-51304-9
2. In vitro Toxicology Systems. Bal-Price, Jennings ISSN 1557-2153,

https://link.springer.com/book/10.1007/978-1-4939-0521-8
3. The Three Rs and the Human Criterion. Micheal Balls. https://replacinganimalresearch.org.uk/the-3rs-and-the-humanity-criterion/

Additional course material will be provided via Canvas

Additional Information Target Audience

mDDS, mBMS, mCHEM, master students in life sciences

Statistics with R

Course Code	X_418156
Credits	6
Period	P2
Course Level	400
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. D. Molenaar
Examiner	dr. D. Molenaar
Teaching Staff	dr. G. Stolf Jeuken, dr. D. Molenaar
Teaching method(s)	Study Group, Computer lab, Lecture

Course Objective

Biologists and biomedical scientists often need to handle and analyse large sets of biological data. A popular tool for such analyses is the open-source statistical programming environment R. R is used intensively in the community of experimental biologists and biomedical scientists, and most newly published data analysis techniques are first available as R-packages.

This course focuses on obtaining the practical skills to perform data handling and analysis tasks from small to large data sets, and to graphically display and interpret the results. Statistical analyses will center on multivariate analysis and causal reasoning.

After completing the course successfully:

- You can use the R-language for reading, wrangling and analysis of data [applying knowledge and insight]
- You can create reports with integrated code, graphs, tables and text [applying knowledge and insight, communication]
- You know techniques of data stewardship and reproducible research [knowledge and insight]
- You can apply basic concepts of probability theory to problems in data analysis and bioinformatics [applying knowledge and insight]
- You can apply linear modeling and ANOVA or model selection criteria like BIC to investigate statistical relations between variables [applying knowledge and insight]
- You can apply logistic regression, naïve Bayes and decision trees to build classifiers [applying knowledge and insight]
- You can interpret and draw causal graphs and infer statistical relations between variables from causal relations [judgement, applying knowledge and insight]
- You can critically evaluate the role of data analysis in the experimental cycle [judgement]
- You can apply linear and other modeling techniques to perform causal inference [applying knowledge and insight]

Course Content

The first part of the course will focus on training the technical aspects of the R-language and of tools for reproducible working.

- Programming effectively in the R language, knowing the basic data structures and their relations, constructing graphs using *ggplot* and apply data manipulation techniques using *tidyverse* functions.
- Document generation with integrated code, cross references and literature citations, and techniques for reproducible calculation.

In the second part we will apply R in probability, statistical modelling and machine learning using examples from various fields of biology, ranging from cell biology to ecology.

- Probability theory: applying the basic concepts of probability theory: sample and event spaces, axioms and

theorems of probability, random variables, probability distributions, conditional and joint probability, dependence, Bayes rule, expected values.
- Statistical modeling: create and analyse statistical models, and knowing its basic concepts: linear modeling, techniques of model selection and model comparison like ANOVA, cross validation, information criteria.
- Causal inference: knowing how to interpret and construct causal diagrams for a data generating process, and knowing how to perform causal inference from causal diagrams and data.

Additional Information Teaching Methods

On campus lectures, work groups, none being mandatory.

Method of Assessment

- Assessment will take place by means of two individual assignments as well as a written exam. Note that these assignments take place on fixed dates and each take approximately a working day to complete. The dates will be announced well before the start of the course.
- To pass the course, the average of the weighted scores on the assignments must be 5.5 or higher and the score on the exam must be 5.5 or higher. The final grade is determined by the scores on the assignments (70%) and the exam (30%).
- Mode of re-examination: one assignment on a similar subject and with the same weight as the last assignment can be submitted once. Also, a re-take of the exam takes place once a year. In all cases the last obtained "pass" score per assessment component (assignments or exam) will be used to calculate the final grade.

Entry Requirements

Bachelor in any science discipline (including medicine). Knowledge of statistical hypothesis testing is required. Some experience with programming is highly recommended. When in doubt please contact the coordinator.

Literature

A course syllabus with theory and exercises is available online.

Additional Information Target Audience

- Master Bioinformatics and Systems Biology
- Master Biomolecular Sciences

Recommended background knowledge

Knowledge of basic statistics is essential. Some experience with a programming language is highly desirable.

Structural Bioinformatics

Course Code	X_405019
Credits	6
Period	P4
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	R. Kumar
Examiner	dr. ir. K.A. Feenstra
Teaching Staff	R. Kumar, dr. ir. K.A. Feenstra, dr. J.M. Urtasun Elizari, dr. H. Mouhib
Teaching method(s)	Computer lab, Written partial exam, Lecture

Course Objective

Why Structural Bioinformatics?

The function of a protein is determined by its three-dimensional structure, and therefore structural information is crucial for understanding the working of proteins. However, experiments, prediction and simulation of protein structures remain difficult. This course will provide you with an overview of existing computational techniques, to validate, simulate, predict and analyse protein structures. More importantly, it will provide practical knowledge about how and when to use such techniques.

After completing the course successfully,

- You understand the basics of statistical thermodynamics and the relation to protein structures, evolution and function [Knowledge and understanding]
- You are able to evaluate protein structures with knowledge of their experimental source, bias and validation techniques [Applying knowledge and understanding]
- You are able to compare different protein structures, and evaluate similarity [Applying knowledge and understanding]
- You know how and when to use structure prediction methods [Making judgements; Applying knowledge and understanding]
- You can perform calculations on protein structures to gain new biological insights [Applying knowledge and understanding]
- You are able to compare different simulation techniques for biological macro-molecules, and be able to analyse the simulated data computationally [Applying knowledge and understanding]
- You are able to read and understand scientific papers in the field of Structural Bioinformatics [Communication; Lifelong learning skills]
- You are able to independently search for answers to open problems and ask a question about something that is unclear [Communication, Lifelong learning skills]

Course Content

Theory:

- Protein and DNA structure sources
- Experimental methods
- Structure validation
- Protein fold prediction (from homology modelling to ab initio prediction)
- Structural classification and structural alignment
- Protein folding and energetics
- Molecular Dynamics & Monte Carlo simulation
- Function from structure

Practical:

- Obtaining geometric features from PDB files
- Homology modelling with Modeller
- Protein interaction as a 'computational experiment' (simulation)

Additional Information Teaching Methods

- 13 lectures (2 two-hour lectures per week)
- Exercises during/in between lectures
- 12 computer practicals (2 two-hour sessions per week)
- Feedback (theoretical and practical) will be given during the computer practical sessions.

Method of Assessment

The final grade for this course will consist of 50% practical work and 50% theoretical assessment.

Practical Assignments: (50%)
(assessment of practical assignments will be adjusted based on ongoing developments in generative language models)

Theoretical: (50%)

- Oral, written or digital exam (depending on number of course students) to assess lecture topics.
- As part of the exam research papers on Structural Bioinformatics topics need to be analysed in detail.
- You will be prepared for your exam through exercises and paper discussions during the lectures, as well as through the practical assignments.

Assessment and grading details will be posted on Canvas.

Entry Requirements

Bachelor in any science discipline (including medicine), with an interest in molecular biology and applying algorithmic approaches to protein structures.
Experience with programming (preferably python).
Basic understanding of calculus and of organic chemistry.
When in doubt please contact the coordinator.

Literature

- course material on canvas.vu.nl
- book: Marketa Zvelebil and Jeremy O. Baum. Understanding Bioinformatics. Garland Science 2008 ISBN-10: 0-8153-4024-9
- book "Introduction to Structural Bioinformatics": The intro is published on ArXiv [https://arxiv.org/abs/1801.09442], and the chapters are available from the table of contents. The book will also be posted on canvas.

Additional Information Target Audience

M Bioinformatics and Systems Biology
M Biomolecular Sciences
M Artificial Intelligence
M Computational Science

Additional Information

BYOD policy (Bring Your Own Device)
We expect students in this course to use their own laptop. This laptop should at the very least support an SSH client, for remote shell access to the VU Linux servers. We will provide some help in setting this up at the start of the course.
Lectures may be recorded; these recordings will only be used within the course, including possibly in subsequent years as background information.

Recommended background knowledge

Some experience with the linux command line is preferred.

The Human Microbiome in Health and Disease

Course Code	AM_1310
Credits	6
Period	P3
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	prof. dr. ir. R. Kort
Examiner	prof. dr. ir. R. Kort
Teaching Staff	dr. D. Molenaar
Teaching method(s)	Seminar

Course Objective

After completing this four-week course (January, 2026), students will be able to:

1. Describe the structure, composition, and function of the human microbiome in different body sites (gut, vagina, mouth, skin, urogenital tract) and explain its role in human health and disease.

Demonstrated through: active participation in lectures and discussions, application in the research proposal
Timeframe/tools: during weeks 1–2, based on lecture material, provided literature and case studies

2.Interpret and critically evaluate microbiome research data and literature, including study design, methodologies (e.g. sequencing, metabolomics), and analysis approaches (e.g. correlation vs causation, AI-based methods).

Demonstrated through: incorporation of peer-reviewed references and methodological reasoning in the written proposal
Timeframe/tools: weeks 1–3, using online databases, lecture content, and feedback sessions

3. Identify potential therapeutic, diagnostic or preventive applications of microbiome science in medical, industrial or environmental domains (e.g. probiotics, dietary interventions, microbiome-based therapies).

Demonstrated through: development of an innovative and feasible microbiome-based application in the proposal
Timeframe/tools: weeks 2–4, using insights from guest lectures and scientific literature

4. Formulate a scientifically sound and original research proposal (4–5 pages) that addresses a microbiome-related problem, includes a clear research aim, an appropriate experimental or analytical approach, a proposed application, and the expected outcomes.

Demonstrated through: submission of a written proposal with abstract, problem definition, approach, application/expected outcome figures (1-3) , references (max. 15)
Timeframe/tools: proposal development in weeks 2–4, using literature, lecture input, and feedback session

5.Communicate research ideas effectively in both written and oral formats, tailored to a scientifically literate audience.

Demonstrated through: 5-minute oral pitch (5 slides), followed by Q&A; clarity of written language in proposal
Timeframe/tools: pitch prepared in the beginning of final week and presented during the final session

6.Engage in constructive peer and instructor feedback to refine scientific ideas and improve research design and communication.

Demonstrated through: active participation in mandatory plenary feedback sessions on January 16 and 23
Timeframe/tools: group sessions with peer/instructor input

7.Reflect on the societal and environmental relevance of the microbiome, including links to planetary health and climate change.

Demonstrated through: integration of these themes in discussion and, where relevant, in the proposal
Timeframe/tools: Planetarium session in the final week; lecture content; optional inclusion in proposal

Course Content

This course provides an in-depth exploration of the human microbiome and its role in health and disease. Students will acquire advanced knowledge about the communities of microorganisms that inhabit the human body, with a focus on their diversity, function, and interactions with the host. The course covers key themes such as the composition of the microbiome in different body sites (e.g. gut, vagina, mouth), the metabolic and immunological roles of microbes, and the impact of lifestyle, diet, and medical interventions on the human microbiome. Major theories and concepts discussed include host-microbe symbiosis, microbial ecology, dysbiosis, the gut-brain axis, and cause-versus-correlation in microbiome studies. Scientific research is at the core of this course. Lectures are given by researchers from academia and industry, and students will critically engage with current literature and methodologies. The course emphasizes applications of the microbiome sciences through the development of a research proposal, in which students design an innovative microbiome-based intervention or study, connecting scientific insights to challenges in health, diagnostics, food, and environmental sustainability. The course concludes with a pitch of the students' proposals and the awarding of the anual ARTIS-Micropia Microbiome Award for the most promising application of the microbiome sciences.

* EXAMPLE OF THE COURSE PROGRAM 2025 * (subject to change in 2026!!)
THE HUMAN MICROBIOME IN HEALTH AND DISEASE 2022 (JAN 7 - JAN 30, 2025)
VENUES: THE LECTURE HALL IN THE VOLHARDING AND THE PLANETARIUM OF ARTIS

WEEK 1

Tuesday, January 7 (10.00 – 12.00 AM) - intro & applications

- An introduction into the human microbiome (10.00 – 10.50 a.m.)
- Prof Remco Kort, A-LIFE, Vrije Universiteit Amsterdam and ARTIS-Micropia
- The holomicrobiome initiative (11.10 – 12.00 a.m.) - Prof Bas Teusink, A-LIFE, Vrije Universiteit Amsterdam
- Wednesday, January 8 (10.00 – 12.00 AM) – the vaginal microbiome
- The vaginal microbiome in health and disease (10.00 – 10.50 a.m.). Prof Remco Kort
- Interactions at the mucosal surface (11.10 - 12.00 a.m.). Dr. Karin StrijbisLinks to an external site., University of Utrecht
- Thursday, January 9 (10.00 – 12.00 AM) – fibers and the gut
- The gorilla gut microbiome. (10.00 – 10.50 a.m.). Prof Remco Kort
- Modulation of inflammation with gut microbiome targeted dietary intervention [recorded lecture]. (11.10 – 11.40 a.m.) Prof Justin SonnenburgLinks to an external site., Stanford University, USA.
- Gut health enhancement by eating favorable food: The GEEF study (11.40 – 12.00). Prof Remco Kort.

WEEK 2

Tuesday, January 14, (10.00 – 12.00 AM) – the gut microbiome

- The human gut microbiome 10.00 – 10.50 a.m. Dr Guus RoeselersLinks to an external site., Danone Research, Utrecht.

- Application of AI tools for the gut microbiome 11.10 -12.00 a.m.. Dr Evgeni LevinLinks to an external site., HORAIZON, Delft
- Wednesday, January 15, (10.00 – 12.00 AM) – cause or correlation?
- Correlation implies causation (but not the way you think it does) (10.00 – 10.50 a.m.). Dr Douwe Molenaar. A-life. Vrije Universiteit Amsterdam
- Inferring cause and effect from observational studies (11.10 – 12.00 a.m.). Dr Douwe Molenaar
- Thursday, January 16 (10.00 – 12.00 AM) – Research proposal session I
- Research proposal plenary feedback session I 10.00 – 12.00 AM. Prof Remco Kort

WEEK 3

Tuesday, January 21, (10.00 – 12.00 AM) – probiotics

- Probiotics: can they deliver what they promise? (10.00 – 10.50 a.m.). Prof. Bruno PotLinks to an external site., Vrije Universiteit Brussel.
- Host-microbe interactions at the mucosal surface (11.10 – 12.00 a.m.). Prof. Ger RijkersLinks to an external site., Professor, University College Roosevelt, Middelburg
- Wednesday, January 22, (10.00 – 12.00 AM) – ecology of microbial communities
- How interactions of microbial communities drive dynamics and function (10.00 – 10.50 a.m.). Dr Matti GralkaLinks to an external site.. A-LIFE. Vrije Universiteit Amsterdam.
- Eco-evolutionary interactions in the urobiome - drivers of disease and antimicrobial resistance (11.10 -12.00 a.m). Dr Marjon de Vos. Faculty of Science and Engineering. University of Groningen
- Thursday, January 23, (10.00 – 12.00 am) – Research Proposal session II
- Research proposal plenary feedback session II Prof. Remco Kort
- Sunday January 26 (24.00 pm), Deadline Research Proposal + Pitch (5 Slides)

WEEK 4

Tuesday, January 28 (10.00 – 12.00 am) – the mouth and the brain

- The oral microbiome in health and disease (10.00 – 10.50 a.m.). Prof. Egija Zaura Links to an external site. ACTA Dental Education, Amsterdam, The Netherlands
- The gut-brain-axis (11.10 – 12.00 a.m.). Prof. Sahar El AidyLinks to an external site., Faculty of Science, University of Amsterdam
- Wednesday January 29 (10.00 – 12.00 am) - Planetarium Session
- 10.00 – 10.50 Our Microbes, our Planet (Micro-organisms and Climate Change) - Prof Jef HuismanLinks to an external site., University of Amsterdam
- 11.10 – 12.00 From Milky way to Microbes – Planetarium dome performance - Milo GrootjenLinks to an external site., Head of ARTIS-Planetarium
- Thursday January 30 The Microbiome Award (sponsored by Yakult)
- Student presentations 10.00 – 12.00 [5 x (5 min presentation + 10 min Q&A)]
- Microbiome Award in ARTIS MICROPIA 12.00 p.m.
- MICROPIA VISIT (all students) 13.00 – 15.00 p.m.

Additional Information Teaching Methods

1. Lectures

Description: Interactive lectures covering the human microbiome, methodologies, applications, and societal relevance. Contact hours per week: ±4 hours (Weeks 1–3). Attendance: **Mandatory.** Details: Includes guest lectures by academic and industry experts; provides the foundation for proposal development and discussions.

2. Literature and Case Study Discussions

Description: Group-based discussions of selected scientific papers and real-world microbiome cases. Contact hours per week: ±2 hours (Weeks 1–2). Attendance: **Mandatory.** Details: Pre-reading required; focuses on interpretation and critical thinking.

3. Feedback Sessions (Plenary & Peer)

Description: Structured feedback moments to improve research proposals through peer and input of Prof Kort. Contact hours per week: ±2–3 hours (Weeks 2–4). Attendance: **Mandatory** (plenary sessions). Details: Includes formative assessment and collaborative refinement of ideas.

4. Oral Pitch Preparation and Presentation

Description: Students prepare and present a 5-minute pitch based on their proposal. Contact hours per week: ±1 hour (Final week). Attendance: **Mandatory** (for own and peers' presentations. Details: Includes Q&A and peer feedback.

5. Planetarium Session

Description: Interactive session linking microbiome science to planetary health and climate change. Contact hours: 2 hours (Final week). Attendance: **Mandatory**. Details: Stimulates reflection on the broader societal and environmental impact of microbiome research.

6. Self-Study / Independent Work

Description: Time allocated for reading, literature search, and proposal development. Contact hours: Not formally scheduled, but students are expected to invest ±10–12 hours per week. Attendance: Not applicable. Details: Essential for successful completion of written and oral course components.

Method of Assessment

1. Written Research Proposal (70%)

Description: Students submit an original and scientifically sound microbiome-related research proposal (4–5 pages), including abstract, problem statement, research aim, methodology, expected outcomes, proposed application, and references (max. 15).

Assessment criteria:

- Scientific quality and originality of the research idea
- Clarity and feasibility of the research aim and approach
- Justification of methodological choices
- Integration of literature and understanding of the microbiome
- Quality of writing and structure
- Feasibility of the application
- Individual / Group: Individual
- Assesses Learning Outcomes: 1, 2, 3, 4, 5, 7

2. Oral Pitch Presentation (20%)

Description: A 5-minute individual oral pitch (with max. 5 slides), followed by a short Q&A session. Students present their proposal to peers and instructors.

Assessment criteria:

- Clarity and persuasiveness of the presentation
- Ability to communicate complex scientific ideas effectively
- Response to questions and interaction during Q&A
- Individual / Group: Individual
- Assesses Learning Outcomes: 5, 6

3. Participation and Engagement (10%)

Description: Active participation in lectures, discussions, feedback sessions, and the planetarium session.

Assessment criteria:

- Constructive contributions to group discussions and feedback
- Demonstrated engagement with reading material and case studies
- Willingness to integrate feedback into research design
- Individual / Group: Individual
- Assesses Learning Outcomes: 1, 2, 6, 7

Entry Requirements

No specific prior course, but a basic understanding of molecular biology and microbiology is required

Literature

A collection of selected publications will be made available on the Canvas site.

Additional Information Target Audience

The target audience includes but is not limited to MSc students Biomolecular Sciences and Biomedical Sciences

Additional Information

Venue: The lecture room in 'de Volharding' in 'ARTIS' (Plantage Kerklaan 38-40, 1018 CZ Amsterdam)

Recommended background knowledge

Recommended background knowledge

General and Molecular Microbiology

Topics in Biomolecular Sciences

Course Code	AM_1311
Credits	6
Period	Ac. Year (September)
Course Level	500
Language Of Tuition	English
Faculty	Faculty of Science
Course Coordinator	dr. ir. Y.J.M. Bollen
Examiner	dr. ir. Y.J.M. Bollen
Teaching Staff	dr. M. Gralka, dr. D. Bald, dr. ir. Y.J.M. Bollen, dr. S. Luirink, dr. J.R. Haanstra, dr. E. Tutucci, prof. dr. ir. E.J.G. Peterman, dr. J.P. van Ulsen
Teaching method(s)	Digital partial exam

Course Objective

The aim of this course is that the student can independently study a relevant topic in biomolecular sciences from scientific literature, and that the student can discuss this topic at the level of a scientific expert.

Course Content

Several pre-defined topics are available of which you can choose one. An expert lecturer will provide a description of the topic, the form of assessment and (starting) literature via Canvas. The student will study the topic and complete the assignment.

Alternatively, the student can also write a literature review on a topic of choice, if (s)he can find an expert supervisor to guide the writing process and grade the review.

Additional Information Teaching Methods

Mostly self-study. Upon request a meeting with the lecturer can be arranged.

Method of Assessment

The form of examination varies per topic. In any case, the student needs to demonstrate master-level understanding of the topic by being able to present a structured overview of the chosen topic, to analyze relevant problems and suggest solutions or new experiments. If the form of examination is a written review, the use of generative AI is only allowed after explicit concent of the expert lecturer, and a detailed reflection of the prompts used needs to be provided.

Entry Requirements

Mandatory for students in the master Biomolecular Sciences

Literature

The lecturer will provide (starting) literature via Canvas.

Additional Information

Some topics are restricted to specific periods in the academic year due to availability of the respective lecturer, and the number of students that can choose a particular topic may also be limited.